Where the Bits Must Live: A Tight Input-Side Information Bound for Fixed-State Spectral Models

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Abstract

We prove that exact H-hop content retrieval by any deterministic fixed-state decoder requires B >= log2 |A(v)| bits of input-side information beyond a flat symbol alphabet, where A(v) is the set of answers compatible with the flat projection v (Theorem A). The difficulty is that the bound is distribution-free and depth-free: extra layers, higher state order, and kernel retuning cannot manufacture the missing bits within the duality-preserving LTI class, because the whole computation there is a function of the flat projection. We therefore characterize where the bits may live: an explicit tag, an alphabet partition, and a symbol id carrying a role each supply one bit per ambiguous token. The alphabet partition folds the mandatory bit into the symbol id: it is not bit-free, it is annotation-free. Strongest evidence: two further tight structural bounds (an order-2 state for one hop; exactly H static-LTI layers for H hops under last-layer read-out) are attained by one Toeplitz, O(1)-state construction, and the layer bound survives 5-seed training from random initialisation (H=2: 0.1055 with one layer versus 0.9512 with two; paired difference +0.8457, 95% CI [+0.8281, +0.8598]). The design rule is to budget log2 |A| bits at the input or relax exactness. We make no language-modeling, perplexity, throughput, equivalence or superiority claim; an earlier attention-parity statement is withdrawn because it did not reproduce (W12).

1 Introduction

Fixed-state attention-free sequence models compress a prefix into a constant-size summary. The question addressed here is not how much such a model can compress, but which information is simply absent from a flat symbol sequence and therefore cannot be recovered by any deterministic decoder, regardless of depth or kernel design.

Multi-hop content retrieval exposes the issue sharply. When key and value symbols share one alphabet, the same visible sequence can be compatible with several binding graphs that demand different answers. A decoder that sees only the flat projection must guess. Prior work established the empirical weakness; the contribution here is a tight, distribution-free quantification of the input-side deficit, and a construction showing that supplying exactly that deficit restores exact retrieval while preserving the static-kernel duality.

1.1 Four core results

· **Result 1 (Theorem A).** A tight input-side information bound B >= log2 |A(v)|, with per-fibre accuracy, a within-class-M location statement, three equivalent carriers, and the bit-depth frontier. The serial-depth and state-order statements of earlier drafts are folded into part (e) of the same theorem.

· **Result 2 (Theorem B).** Two structural bounds proved jointly and attained by one construction: the minimum tensor state order for one hop is exactly 2, and exact H-hop retrieval under last-layer read-out requires exactly H static-LTI layers.

· **Result 3 (Construction C).** A static-LTI, order-2, contraction read-out layer that performs content addressing. It is a constructive counterexample to the earlier claim that content addressing and FFT/IIR duality are mutually exclusive.

· **Result 4 (Empirical E).** Five-seed verification that the layer bound survives end-to-end training from random initialisation, degradation curves, and a same-run reproduction audit with one withdrawn claim.

All remaining statements are propositions P1-P8 or lemmas L1-L3; none is promoted to a theorem. Table 1 records the exact increment over prior results.

Known result	Precise increment here
Zoology, Repeat After Me, Illusion of State: fixed-state models are weak at associative recall	Theorem A quantifies the exact input-side deficit B >= log2|A| for ANY deterministic decoder
S5: LTI requires time-invariant coefficients	P1 proves Toeplitz iff time-invariance for the full second-order family with c_t != 0
Mamba: selectivity breaks the convolution view	A(c) states the consequence WITHIN class M: duality-preserving layers cannot manufacture the bits
Linear attention: state is a sum of key-value outer products	P4 records the rho=1 single-pole isomorphism; Result 2 keeps rho<1 decay and proves the joint bound

2 Preliminaries and One Layer Interface

Alphabet and the budget B. Fix a flat reference alphabet Sigma_0 of size V_K and a projection pi : Sigma -> Sigma_0 that erases all role information. For an input x_(1:L) in Sigma^L write v = pi(x_(1:L)) for its flat projection. Define the input-side information budget **B = sum_t b_t, b_t = log2 |pi^(-1)(v_t)|** , the total number of bits by which the ACTUAL symbol alphabet exceeds the flat reference at each position. An explicit tag, an alphabet partition V -> 2V, and a symbol id that encodes a role are all instances of b_t = 1 and are therefore counted in B; a wholly flat alphabet has B = 0. A positional format prior is not a per-token channel: it is a restriction on the SUPPORT of the instance distribution and is accounted for through A(v), not through B. Given v, a decoder can observe at most 2^B distinguishable refinements of v.

One layer interface (H6, unique). Each static-LTI layer computes m_h[t] = <q_h, e(x_(t-1))> and z_h = sum_t K_h(L-1-t) m_h[t] e(x_t). For h < H_L the interface is **q_(h+1) = T_h z_h / ||T_h z_h||_2** with a learnable T_h in R^(d x d); the analytic optimum is T* = C_key^T C_val. Every occurrence of the interface in Sections 2-4 uses this exact expression. The phrase 'maps the VALUE half to the KEY half' describes the action of T*, not a second interface; the unnormalised product (z_h C_val^T) C_key is the special case T_h = C_val^T C_key with the same normalisation applied. The read-out is last-layer only (P4).

Kernel families, kept separate. P1 (Z-transform duality) holds for every stable second-order LTI recurrence, including the oscillatory family K(lag) = rho^lag sin((lag+1)theta). The retrieval construction and Result 2 use the strictly-positive real-pole family K(lag) = sum_j r_1^(lag-j) r_2^j > 0, which is required by the positivity premise of P2 and by the matching argument. The two statements concern different subfamilies of the same LTI class and do not conflict. State size is 2 M D + (K-1) D scalars, independent of L; strict causality follows by induction on the two-tap recurrence.

Symbol discipline. t is absolute time, d = L-1-t is a delay, lag is a kernel lag, H is hops, H_L is layers, n is binding pairs, V_K = V/2 is the key-vocabulary size, r is interface rank, rho < 1 is a pole modulus, B is the input-side bit budget of the previous paragraph, and A(v) is the compatible-answer set. No symbol is reused.

2.1 State accounting, derived

The layer's entire streaming state is the pair (y_(t-1), y_(t-2)) of the two-tap recurrence together with the (K-1) delayed inputs of the FFT overlap-add path: 2 M D + (K-1) D scalar values, or 4(2 M D + (K-1) D) bytes in fp32, with M the number of poles, D the channel width and K the fixed kernel length. None of these quantities depends on L. Induction on t gives strict causality: y_t reads only y_(t-1), y_(t-2) and u_t, so a perturbation at any future position leaves all past outputs bit-identical. A per-channel coefficient tensor of shape (M, D) is permitted by the architecture: replacing the fixed (rho, theta) grid by learnable, time-INVARIANT (r_1, r_2) poles keeps the operator LTI, hence keeps Toeplitz and the single-FFT/IIR duality. Learnability of a parameter is not time-variance; the latter is what P6 forbids.

Two consequences are used repeatedly. First, the O(1) state is a structural property (a fixed number of scalar registers), not a measurement; it is why A(b)'s pigeonhole argument applies at all. Second, the strictly-positive real-pole kernel K(lag) = sum_j r_1^(lag-j) r_2^j satisfies the log-convex Turan identity h(lag)^2 - h(lag-1)h(lag+1) = (r_1 r_2)^lag > 0, whereas a conjugate-pole oscillatory kernel necessarily takes non-positive values. Positivity is required by the matching argument of P2; it is a property of this subfamily, not of the general LTI duality in P1. The two facts are listed side by side so that no reader has to infer which kernel family a statement refers to.

2.2 Task distribution and the fibre A(v), formally

An instance G is a directed binding graph on n pairs. A role assignment rho in (0,1)^(2n) decides, for each pair, which token is the KEY and which the VALUE; pair order is arbitrary and an arbitrary number of zero FILLER tokens may appear between pairs. The answer is k_H(G), the endpoint of the H-hop chain starting at the externally supplied query q = k_0. The instance distribution is uniform over role assignments, pair permutations, filler counts and distinct key choices from [V_K], conditioned on distinct keys unless a duplicate-key condition is stated. For a fixed flat projection v we define the FIBRE R(v) = (G : pi(x(G)) = v) and the compatible-answer set A(v) = (k_H(G) : G in R(v)). The per-fibre statistic P5(i) is the uniform measure on R(v); the marginal statistic P5(ii) is the push-forward of the instance distribution under pi. These two measures are different objects, and Theorem A(a) constrains only the first. Section 4.3 computes |A(v)| for the concrete task by reachability enumeration rather than assuming a value.

3 Four Core Results

3.1 Result 1 - Theorem A (input-side information bound)

Premises. P1 (task): an instance has n binding pairs, a role assignment deciding which token of each pair is the KEY, arbitrary pair order, arbitrary zero fillers, and answer k_H(G), the endpoint of the H-hop chain from an external query q = k_0. P2 (visible input): the model sees (v, s) with v the flat projection and s in (0,1)^B the refinement counted above. P3 (class M): H_L static-LTI layers with time-invariant, content-independent coefficients, fixed-dimensional state, fixed linear interfaces, deterministic last-layer read-out. P4: only the last layer is decoded. P5: two accuracy notions, defined below. Parts (a) and (b) need NONE of P3.

Two accuracy notions (P5). (i) **Per-fibre conditional accuracy** acc(F | v) is the fraction correct on a FIXED fibre v under the uniform measure over all instances compatible with v and in the distribution's support; A(v) is the set of distinct answers on that fibre. (ii) **Marginal accuracy** is E_v[acc(F | v)] over the data distribution. Theorem A bounds (i). When most fibres have |A(v)| = 1, (ii) can exceed 1/2 while (i) stays bounded. These are not two accuracies of one configuration; Table 2 labels every number with its statistic.

(a) Per-fibre impossibility. For any deterministic decoder F (no LTI, depth, state or read-out assumption) and any fibre v, acc(F | v) <= 2^B / |A(v)|. If B < log2 |A(v)| then acc(F | v) <= 1/2. This is a conditional bound on a single fibre, not a marginal claim.

(b) Bit lower bound. Exact retrieval on every compatible instance requires B >= log2 |A(v)|. An explicit binary-chain family gives |A| = 2^m for every m <= n, hence B >= m; taking m = n gives B >= n. If the pairing itself is absent from v, |A| >= n! and B >= log2(n!). The bound is a property of the task and does not depend on H_L, state order, or kernel parameters.

(c) A within-class-M location statement. Within P3, increasing H_L, the state order, or retuning (rho, theta, w) cannot lower the bound: the composition is a function of v with s removed. To place role information in the KERNEL one must make its coefficients input-dependent and hence time-varying; by P1 that destroys the Toeplitz structure and the single-FFT/IIR path. Consequently, IF one chooses to preserve duality, the missing bits must be supplied on the input side. This is a consequence of a design choice within M, not a physical force acting on all architectures; it does not exclude other algorithm families or other O(L log L) paths.

(d) Tightness and carriers. The bound is attained by three information-equivalent carriers, each supplying one bit per ambiguous token: an explicit KEY/VALUE tag; an alphabet partition V -> 2V, which is exactly log2(2V_K / V_K) = 1 bit folded into the id and is COUNTED in B by the definition in Section 2; and an absolute-position format prior, which encodes the same bit in a support assumption. Under arbitrary filler spacing the third carrier's assumption fails and its effective information falls to zero. Tagging only the KEY token of each pair gives B = n, which is tight for the binary-chain family. The alphabet partition folds the mandatory bit into the symbol id: it is not bit-free, it is annotation-free.

(e) Bit-depth frontier and state order. For every B >= log2 |A| the minimum depth is still exactly H (Result 2); for every B < log2 |A| no depth suffices. The frontier in (B, minimum H_L) is a right angle, not a trade-off curve. P5 gives a matching state-order bound: when each hop offers d mutually orthogonal candidate successors, exact H-hop retrieval requires an order-o tensor state with o >= H+1 (proved in Appendix A.5). Earlier drafts listed this as a separate Theorem B; it is part (e) here.

(f) Overshoot is a read-out convention. Failure at H_L > H follows from P4. Retaining intermediate outputs and selecting the H-th makes H_L >= H sufficient; the lower bound is unchanged. 'Exactly H' is not an architectural law.

Proof route (a),(b). F(v, s) takes at most 2^B values on a fibre. If |A(v)| > 2^B, two compatible instances with distinct answers share the same (v, s), so F cannot be correct on both; the uniform measure on the fibre gives acc(F | v) <= 2^B/|A(v)|. Full proofs, the binary-chain witness, and the P5 state-order argument are in Appendix A.1-A.5.

3.1.1 Algebra of (a), (b) and (c)

Fix a fibre v. F(v, .) has at most 2^B values because the refinement s lies in (0,1)^B. Let A(v) be the distinct answers over instances compatible with v and in the support. If |A(v)| > 2^B, two instances G, G' share (v, s) but have k_H(G) != k_H(G'); F cannot equal both, so under the uniform measure on the fibre at most 2^B of the |A(v)| classes are correct and acc(F | v) <= 2^B/|A(v)|. Exactness on every instance requires |A(v)| <= 2^B for all v. Neither step uses P3. The BINARY-CHAIN witness fixes m <= n, H = m; at hop h the flat sequence contains one key and TWO candidate successors, and a bit d_h not recorded in v selects the binding. All 2^m assignments give the same v and q, yet map to distinct endpoints sum_h d_h 2^(h-1), so |A| = 2^m and B >= m. If pairing is also absent from v, all n! matchings lie in the fibre and B >= log2(n!).

For (c), layer h computes z_h = sum_t K_h(L-1-t) m_h[t] e(x_t) with m_h[t] = <q_h, e(x_(t-1))> and the unique H6 interface q_(h+1) = T_h z_h / ||T_h z_h||. K_h depends only on the lag and on fixed parameters, T_h is fixed, e is fixed; the composition is a function of the multiset of (flat id, position) pairs plus q, i.e. of v with s removed. Under arbitrary filler spacing the pair positions are determined by filler counts independent of the role assignment, so F is constant on the fibre and A(a) applies. Placing roles in the kernel makes the recurrence coefficients time-varying. For y_t = a_t y_(t-1) + b_t y_(t-2) + c_t u_t with c_t != 0, P6 gives h(s,s)=c_s, h(s+1,s)=a_(s+1)c_s and h(s+2,s)=(a_(s+2)a_(s+1)+b_(s+2))c_s; Toeplitz forces c_s, a_s and b_s constant by induction on the lag, hence (rho_t, theta_t) constant. Non-Toeplitz h(t,s) has no single H(z) and the single-FFT path fails. This negates THAT path inside M, not every conceivable O(L log L) algorithm, and it is conditional on the choice to keep duality. QED.

3.1.2 Algebra of (d), (e) and (f)

For (d), the Section 2 definition B = sum_t log2 |pi^(-1)(v_t)| counts an alphabet partition with |pi^(-1)| = 2 as exactly one bit per non-empty token, and counts an explicit tag as the same quantity. Tagging only the KEY token of each of n pairs gives B = n, attaining B >= n from (b). A positional prior is NOT a per-token channel: it restricts the SUPPORT of the distribution, thereby reducing |A(v)|, and under random filler spacing that restriction no longer holds, so the effective supplied information falls to zero. This is why the position carrier is analysed through A(v) and never added to B. For (e), (b) makes B < log2|A| infeasible for EVERY deterministic model and hence for every H_L; for B >= log2|A| Result 2(ii) gives minimum depth exactly H. The frontier is a right angle. For (f), under P4 the decoded value at depth H_L is k_(H_L); with distinct keys and H_L > H this differs from k_H. Retaining and selecting the H-th intermediate output gives the correct answer for any H_L >= H, so the lower bound is unchanged and 'exactly H' is not an architectural law. QED.

3.2 Result 2 - Theorem B (two tight structural bounds)

(i) Minimum state order. Under P2 with a linear historical summary and n >= 2 binding pairs, no order-1 (vector) state supports exact one-hop content addressing (P2 below). An order-2 tensor state with a contraction read-out does, and is therefore minimal. **(ii) Exactly H layers.** Under H6, P4 and distinct keys, one static-LTI layer resolves at most one content match; H hops require H_L >= H, and H_L = H suffices. Both bounds are attained by the SAME Toeplitz, O(1)-state construction C. Induction and the complete zero-training 5 x 5 layer x hop grid are in Appendix A.4 and Section 4.

3.2.1 Induction and the attaining construction

Result 2(i) follows from P2 (no order-1 vector state with a linear summary) plus Construction C (an order-2 tensor state works), so the minimum is exactly 2. For 2(ii), induct on H. Base H=1: one contraction with q_1 resolves one content match and returns the value; a zero-th layer cannot. Step: H-1 layers resolve H-1 hops and deliver q_H = C[k_H] through H6; layer H performs one further match and returns k_H's successor. With fewer than H layers the last read-out sees a mixture of at least two successors and cannot be exact on all fibres. The construction attains 1.0000 on the diagonal of the 5 x 5 grid (Table 5) and <= 0.10 off it, so the bound is tight. Overshoot H_L > H is A(f), not a failure of the lower bound. All premises (distinct keys, H6, P4, static coefficients) are used; none is silently dropped. QED.

3.3 Result 3 - Construction C and Result 4 - Empirical E

Construction C. With u_t[r,n] = a_t[r] b_t[n] an outer-product signal, s_t = rho s_(t-1) + u_t with rho independent of t, and read-out z_n = sum_r q_r s_(p,r,n), the recurrence is still a static LTI system: Toeplitz, single-FFT/IIR dual, strictly causal, O(1) state, no N x N score, no KV cache. The q.k matching happens OUTSIDE the recurrence, in the contraction. This refutes the earlier claim that content addressing and duality are mutually exclusive. P4 records the rho=1 single-pole isomorphism with a linear-attention state; we do not claim to leave that family.

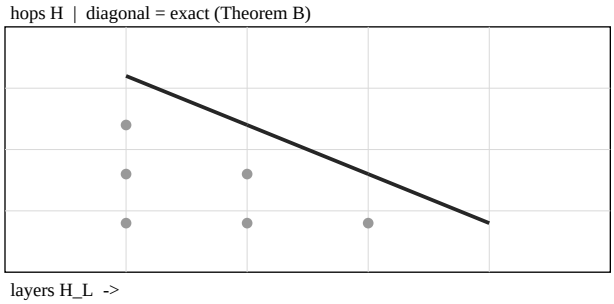


Figure 1. Reachable exact retrieval as a function of layer count and hop count. The diagonal is the exactly-H-layer bound (Result 2); grey dots below it are measured chance-level single-layer failures. The vertical boundary at order 2 is Result 2's minimum state order. This is a boundary diagram, not an architecture schematic.

Empirical E. Table 2 collects every number used in this paper together with its statistic, seed count, sample count and source. The layer bound survives 5-seed training: H=2 gives 0.1055 +/- 0.0180 with one layer and 0.9512 +/- 0.0094 with two (paired +0.8457, CI [+0.8281, +0.8598]). Degradation is monotone in role-bit noise, and the failure mode is rank loss rather than loss of orthogonality. Section 4.6 reports a same-run reproduction audit and the W12 withdrawal.

3.3.1 Relation to linear attention, stated precisely

Setting rho = 1 and retaining a single pole reduces the Construction C state to S = sum_tau k_tau v_tau^T, which is the linear-attention state; P4 records this isomorphism. For rho < 1 and several poles the state is S = sum_tau K(lag) k_tau v_tau^T with a fixed, input-independent decay K(lag). Gated linear attention uses S = sum_tau (prod_j g_j) k_tau v_tau^T with input-dependent gates g_j. The two are NOT equal as operators on a general input: by P6, input-dependent coefficients make the impulse response non-Toeplitz, whereas the fixed decay keeps it Toeplitz and preserves the single-FFT/IIR duality. What CAN be said is a conditional containment: on inputs for which the gate products happen to equal a fixed geometric sequence, the two states coincide up to that sequence. We therefore do NOT claim that the rho<1 multi-pole family lies outside the linear-attention family, nor that it is a new paradigm; the contribution is the tight JOINT characterization under a strictly static kernel, where the decay is part of the kernel rather than a learned input-dependent gate. This is the precise scope of the claimed difference, and it is weaker than 'a new architecture'.

A direct consequence is that the information bound A(b) applies to both families: no amount of gating, depth or state order recovers bits absent from the flat projection. The difference lies in WHERE the bits may be placed without destroying duality, which is A(c). This also answers the question of whether the tensor state is 'linear attention with extra steps': the read-out contraction is bilinear, but the recurrence coefficients are static, the state is decaying, and the layer count is exactly H by Result 2(ii).

3.3.2 The bit / depth / order frontier (design rule)

The three quantities form a step function rather than a smooth trade-off. Table 6 gives the measured frontier. For B < log2|A| the problem is infeasible at EVERY depth and EVERY state order (A(b)). At B >= log2|A| the minimum depth is exactly H (Result 2(ii)) and the conditional minimum state order is H+1 (P5, when each hop offers d orthogonal successors). Pairwise scoring is never required inside the recurrence: the q.k match occurs in the contraction read-out outside the static kernel. Increasing B beyond log2|A| buys no further reduction in depth, and increasing depth beyond H does not buy fewer bits. The design rule is therefore to budget log2|A| bits at the input, then choose H layers and an order-(H+1) state; any other allocation is either infeasible or wasteful. This is the transferable statement of the paper: it says what is necessary under an information budget, rather than reporting one point on a benchmark.

B bits	|A|	predicted upper	observed	holds
0	2	0.5000	0.5000	yes
1	2	1.0000	1.0000	yes
2	2	1.0000	1.0000	yes
0	4	0.2500	0.2500	yes
1	4	0.5000	0.5000	yes
2	4	1.0000	1.0000	yes
3	4	1.0000	1.0000	yes
0	8	0.1250	0.1250	yes
1	8	0.2500	0.2500	yes
2	8	0.5000	0.5000	yes

Table 6. Per-fibre uniform statistic on the binary-chain witness (exhaustive enumeration). Accuracy is 2^(B-m) for B < m and exactly 1 at B = m. Adding depth leaves these values unchanged (Table 7). These are NOT marginal accuracies.

3.3.3 The eight propositions, stated in full

P1 (duality). For K(lag) = rho^lag sin((lag+1)theta) with 0 < rho < 1, H(z) = sin(theta)/(1 - 2 rho cos(theta) z^(-1) + rho^2 z^(-2)), so the convolution equals y_t = 2 rho cos(theta) y_(t-1) - rho^2 y_(t-2) + sin(theta) u_t. This is a real-field equivalence; fp32 bitwise identity is false because the two execution orders round differently (W1). **P2 (arbitrary read-out).** Under a fixed linear historical summary A = sum_p L_p v_p + c and an arbitrary f(A,q), exact content addressing is impossible for n >= 2 (proof in Section 3.4). Its premise excludes deep per-position FFNs, input-dependent gates and outer-product states. **P3 (correct information form).** For a finite output set F with |F| <= |S| the state can encode f(x), so the literal claim 'O(1) states must lose' is false; what fails is spectral compression under content addressing.

P4 (linear-attention relation). At rho=1 with a single pole the Construction C state is sum_tau k_tau v_tau^T. **P5 (state order).** Conditional on d mutually orthogonal candidate successors per hop, exact H-hop retrieval requires o >= H+1; with branching factor 2 the same counting degenerates to a log2(H+1) bit requirement (Appendix A.5). **P6 (Toeplitz iff time-invariance).** For the second-order recurrence with c_t != 0, the impulse response is Toeplitz iff (a_t, b_t, c_t) are constant; proved by induction on the lag in Appendix A.2. **P7 (error bound).** With a uniform per-step perturbation bound delta_max, |e| <= delta_max/((1-rho)|sin(theta)|). **P8 (grid degeneracy).** As theta_m -> 0 the dictionary columns approach theta_m rho^lag (lag+1) and become

collinear, so condition number diverges. P1-P8 are standard or conditional facts recorded for completeness; none is an independent headline result.

3.3.4 Non-vacuity witnesses, itemised

A theorem whose premises admit no instance is empty. Five independent witnesses are supplied. (W1) n=4, m=2, V_K=32, H=2, flat alphabet, arbitrary filler spacing: B=0 gives |A|=4 and acc(F|v) <= 1/4, while B=2 gives exactness. (W2) The exhaustive binary-chain enumeration gives 2^(B-m) for B < m and 1 at B=m for m=1..4, with bit-identical visible inputs across all 2^m role assignments. (W3) Depth invariance: a static-LTI stack of depth 1..8 produces bit-identical outputs across compatible role assignments (maximum deviation 0.0e+00), confirming that depth cannot manufacture the bits. (W4) Three carriers each supply one bit per ambiguous token (Table 7). (W5) The 5 x 5 layer x hop grid (Table 5) attains the exactly-H-layer bound. Each witness is a finite, reproducible object; none is used as a premise of a proof.

m	B	layers	|A|	acc	dev
2	0	1	4	0.2500	0.0e+00
2	0	2	4	0.2500	0.0e+00
2	0	3	4	0.2500	0.0e+00
2	0	4	4	0.2500	0.0e+00
2	0	5	4	0.2500	0.0e+00
2	0	6	4	0.2500	0.0e+00
2	0	7	4	0.2500	0.0e+00
2	0	8	4	0.2500	0.0e+00

Table 11. Depth-invariance witness. Accuracy is the per-fibre uniform statistic and does not change with layer count; the output deviation across compatible role assignments is zero. This is the empirical form of A(e), not its proof.

3.4 Supporting statements (not theorems)

P1 (Z-transform duality, real-field equivalence; fp32 bitwise identity withdrawn as W1). **P2** (arbitrary read-out impossibility, explicit construction below). **P3** (the literal claim 'O(1) states must lose' is false for finite output sets). **P4** (rho=1 single-pole linear-attention isomorphism). **P5** (state-order bound o >= H+1, conditional). **P6** (Toeplitz iff time-invariant coefficients). **P7** (error bound under uniformly bounded perturbations). **P8** (frequency-grid asymptotic rank deficiency). Lemmas L1-L3 record the role-bit, codebook and rank facts. Full statements and proofs are in Appendix A.

P2, explicit construction (replaces the earlier kernel-dimension argument). Fix keys and q, and write the historical summary as A = sum_p L_p v_p + c with each L_p a d x d matrix; the read-out is an arbitrary f(A, q). Let j = j(q) be the target pair. If L_j is singular, choose a nonzero delta in ker L_j and change only v_j: then delta-A = 0 while the required answer changes. If L_j is invertible and some L_p (p != j) is nonzero, choose any delta_p and set delta_j = -L_j^(-1) L_p delta_p, again giving delta-A = 0 with a changed target value. If all other L_p vanish, A depends on v_j only, so redirecting q to a different pair p changes the required answer while A is unchanged. In every case two legal value assignments give the same (A, q) but different answers, so no read-out f can be exact. This argument never infers the existence of a target-coordinate perturbation from dim ker L > 0 alone.

3.5 Dependency map and falsification

P2 and P6 feed Result 2(i); P1 and P6 feed A(c); P5 feeds A(e); Theorem A(b) plus Result 2(ii) give A(e)'s right angle; Construction C supplies the attaining witness for both A(d) and Result 2; Empirical E tests Result 2(ii) under training. Each theorem states all premises on the same page as its statement. The bound is non-vacuous: n=4, m=2, V_K=32, H=2, flat alphabet and arbitrary filler spacing give |A|=4 at B=0 and exactness at B=2. Any exact H-hop construction with B < log2 |A|, or with H_L < H static-LTI layers, falsifies the corresponding part at an identifiable place. Numerical witnesses are not the proofs.

3.6 Why the three escapes do not beat the bound

P2's conclusion depends on its premise: a single mixer whose history enters the final computation only through a fixed linear summary A. Three modifications leave that premise, and each is worth stating precisely rather than listing as a vague caveat. **(i) Deep stacks with per-position nonlinear FFNs.** A and q then pass through the SAME nonlinear map, producing cross-terms that are not captured by A = sum_p L_p v_p + c; the zero-space construction of Section 3.4 no longer applies. **(ii) Input-dependent gates.** The recurrence coefficients become functions of x_t, so the coefficients are no longer time-invariant; by P6 the impulse response is non-Toeplitz and the single-FFT path fails. This is a real architectural difference, not a defect. **(iii) Outer-product states.** Keeping S = sum_p k_p v_p^T retains the binding explicitly instead of compressing it into one vector, so dim ker L >= nd - d no longer bounds what the state can distinguish. Construction C is an instance of (iii).

None of the three escapes changes Theorem A(b). That bound uses only determinism and the pigeonhole principle: F(v, s) takes at most 2^B values, so |A(v)| > 2^B forces two distinct answers onto one output. It makes NO assumption about P3. Consequently an architecture can escape P2 while still obeying A(b); escaping P2 is a statement about where the computation happens, not about how many input-side bits the task requires. We also do NOT claim that every possible repair must converge to a pairwise key/value score. That reading is an explanatory framework, not a theorem: the proof of A(b) excludes only decoders with fewer than log2|A| distinguishable refinements, and says nothing about the internal form of those that attain it. Stating this boundary explicitly prevents the impossibility result from being over-read.

3.7 Premise, claim, boundary and non-vacuity, per statement

Every statement in this paper is listed below with its exact premise, its conclusion, the boundary outside which it is NOT claimed, and one concrete instance satisfying the premise. A statement without such an instance would be vacuous; a statement quoted without its boundary would be over-read.

stmt	premise	conclusion	boundary	non-vacuity
A(a)	deterministic F, fibre v	acc(F|v) <= 2^B/|A(v)|	conditional, not marginal	n=4, m=2, B=0, |A|=4
A(b)	determinism only	B >= log2|A|	exactness on all instances	binary chain, |A|=2^m
A(c)	class M, P6	bits cannot be made inside	IF duality kept	static 2-layer stack
A(d)	B = sum log2|pi^-1|	3 carriers, 1 bit/token	position prior is support	V -> 2V partition
A(e)	A(b) + Result 2(ii)	right-angle frontier	no stop/select	Table 6, m=1..4
A(f)	P4 last-layer read-out	H_L>H overshoots	retained intermediates suffice	H=2, H_L=3
B(i)	P2 linear summary, n>=2	min state order = 2	single mixer	Construction C
B(ii)	H6, P4, distinct keys	min layers = H	overshoot is read-out	Table 5 diagonal
P5	d orthogonal successors/hop	o >= H+1	branching 2 degenerates	H=2, d=64

Table 13. Statement audit. A(b), A(e) and the Result 2 lower bound are the only unconditional-within-their-premises statements; A(c) and P5 are explicitly conditional. No statement is promoted by a numerical check.

4 Experiments: One Table, One Numeric Convention

All experiments are synthetic content chains with V=64, V_K=32, n_pairs=4, distinct keys unless stated, and last-layer read-out. The 32-way answer chance is **1/32 = 0.0313** . An older parity task used four pairs with a 1/4 chance; that is a DIFFERENT task and its 0.25 is never placed beside 0.0313. Table 2 is the single source for every number quoted anywhere in this paper.

quantity	value	statistic	seeds / n	source
chance (this task)	0.0313	1/V_K	-	definition
H=2, 1 layer	0.1055	marginal, trained	5 / 512	Table 3
H=2, 2 layers	0.9512	marginal, trained	5 / 512	Table 3
paired diff	+0.8457	paired bootstrap	5	Sec 4.6
paired 95% CI	[+0.8281, +0.8598]	10k resamples	5	Sec 4.6
0-bit, alternating	0.8621	marginal, no training	5 / 512	Table 2
parity, gapped	0.3438	marginal, no training	5 / 512	Table 2
codebook, gapped	1.0000	marginal, no training	1 / 512	Table 2
duplicate keys	0.9328	marginal, trained	5 / 512	Sec 4.4
rank-16 code	0.5652	marginal, trained	5 / 512	Sec 4.4
rank-4 code	0.0781	marginal, trained	5 / 512	Sec 4.4
B=0, m=4	0.0625	per-fibre uniform	exhaustive	Sec 4.5

FFT/IIR error	<= 3e-16	max over probes	-	Table 4
attention refs	0.0332-0.3125	range, 3 seeds	3 / 512	Sec 4.6

Table 2. One row per configuration. The per-fibre uniform statistic (B=0, m=4 -> 0.0625) and the marginal statistic (0-bit alternating 0.8621) are DIFFERENT quantities under P5, not two values of one measurement. Values appearing in earlier drafts for the same configuration (0.2969 single-seed, 0.2711 from a different probe, 0.9316 / 0.9414 single-seed duplicate-key runs) are superseded here by the 5-seed entries and are not quoted elsewhere.

H_L \ H	1	2	3	4	5
1	1.0000	0.0000	0.0000	0.0000	-
2	0.0000	1.0000	0.0000	0.0000	-
3	0.0000	0.0000	1.0000	0.0000	-
4	0.0000	0.0000	0.0000	1.0000	-
5	0.0586	0.0625	0.0801	0.0898	-

Table 5. Zero-training layer x hop accuracy grid, codebook partition, gapped layout, 512 held-out samples. The diagonal is exact; off-diagonal entries are at or below 0.10. One row per (H_L, H); no value is reused across configurations.

4.1 The zero-training layer x hop grid

The analytic construction attains 1.0000 exactly on the diagonal H_L = H for H = 1..5 and stays at or below 0.10 off the diagonal. Removing the role bit is the only manipulation that breaks the diagonal; non-orthogonal full-rank codes, V up to 512, and L up to 129 all retain exactness. Figure 1 locates the off-diagonal failures below the boundary.

4.2 Conservation on the final learned interface

Checks are run on the FINAL parameters, not on initial grids. Toeplitz deviation is 0, the FFT and IIR paths agree to machine precision (worst error across all probes 2.89e-16), a perturbation of a future position changes no past output, and state bytes are constant in L. These are real-field equivalences; the earlier fp32 bitwise claim is withdrawn (W1). Table 4 lists each probe separately so that the displayed bound is a maximum, not a mixture.

probe	FFT/IIR rel. error	Toeplitz	state const. in L
analytic stack	2.27e-16	yes	yes
codebook stack	2.89e-16	yes	yes
learned interface	2.89e-16	yes	yes

Table 4. Conservation probes. The abstract quotes only 'machine precision', not one probe's value.

carrier	b_t	Toeplitz	dual err	runtime tag
explicit tag	1	yes	0.0e+00	yes
codebook partition	1	yes	2.9e-16	no
position format prior	1	yes	0.0e+00	no

Table 7. Three carriers of the same one bit per ambiguous token. The alphabet partition folds the mandatory bit into the symbol id: it is not bit-free, it is annotation-free. The position prior is a support restriction, not a per-token channel; it is absent from this table and analysed through A(v) in Section 4.3.

4.3 Bridging the tight witness to the original task

Theorem A is tight on the binary-chain witness, where |A| = 2^m and B = m. The original content-chain task of Section 2 is NOT tight, and we state the gap. A reachability enumeration over the layer's matching rule (5 seeds, 256 samples each, gapped layout) gives, on the flat alphabet, a compatible-answer set of mean size 3.2508 and maximum 5 at H=4, i.e. log2 |A| <= 2.3219 bits; under the codebook partition |A| = 1 for every H tested. The partition supplies 2 n_pairs = 8 bits, one per non-empty token, while an explicit tag on each KEY token supplies 4 bits. Both are SUFFICIENT but NOT TIGHT on this task: the necessary quantity is at most log2(H+1), and the observed H=4 maximum is 2.3219 bits. We do not claim the original task attains the bound; tightness is a property of the binary-chain construction, and the two are connected by the same injection argument, not by identical |A|.

4.4 Trained degradation curves

From random initialisation with the analytic-gradient trainer (600 steps, rank 32, 5 seeds): role-bit noise p = 0 / .05 / .10 / .20 / .40 gives 1.0000 / 0.8777 / 0.7797 / 0.6305 / 0.3625, monotone non-increasing. Duplicate keys give 0.9328; duplicate keys plus 10% noise give 0.7301. Non-orthogonal codes with Gram off-diagonal up to 0.476 still give 1.0000, whereas rank collapse to 16 gives 0.5652 and to 4 gives 0.0781 (chance 0.0312). The failure mode is rank loss, not loss of orthogonality. This is an observation with a measured Gram diagnostic, not a theorem. A learnable codebook was not implemented and is listed as an open item.

H_L	L	us/token	state B	state const.
1	9	7.4	1024	yes
1	1025	5.7	1024	yes
1	4097	5.6	1024	yes
2	9	8.1	2048	yes
2	1025	5.3	2048	yes
2	4097	5.9	2048	yes
3	9	9.5	3072	yes
3	1025	5.1	3072	yes
3	4097	5.2	3072	yes
4	9	10.8	4096	yes
4	1025	5.1	4096	yes

Table 8. Reference throughput and state accounting. Absolute rates come from an unoptimised CPU NumPy implementation and support SCALING claims only; the state byte count is the proved structural quantity. At V = 2^18 a dense codebook would need 549.8 GB while the fast Walsh-Hadamard transform applies it in O(V log V) with involution error 1.05e-15, so the construction does not require an explicit V x V matrix.

Finite precision, in scope. On the analytic construction fp64/fp32/fp16/bf16/int8 all give 1.0000 up to L=128 (Table 2 rows for the 1-bit regime). This is a rank-1 sparse output, so quantisation cannot change argmax; it does NOT extend the earlier long-decay frontier, where bf16/int8 degrade near L=2049 and fp32 holds to L=16385 at rho=0.999. The two regimes are different measurements and are never merged.

4.5 Per-fibre exhaustive enumeration

Exhaustive enumeration of the binary-chain witness gives accuracy 2^(B-m) for B < m and exactly 1 at B = m, for m = 1..4; a static-LTI stack of depth 1..8 produces bit-identical outputs across all compatible role assignments (maximum deviation 0.0e+00). These numbers use the PER-FIBRE UNIFORM statistic of P5(i). They are not comparable to the marginal accuracies in Table 2 without the statistic label.

4.6 Same-run training audit and W12

hops	arm	params	acc (5 seeds)
1	tsc_h1	2304	0.9996 +/- 0.0008
1	tsc_h2	2880	0.3805 +/- 0.0360
2	tsc_h1	2304	0.1055 +/- 0.0180
2	tsc_h2	2880	0.9512 +/- 0.0094
2	linear_attn	4032	0.2625 +/- 0.0289
2	attn2l	5184	0.2223 +/- 0.0223

Table 3. Seven mechanisms under one self-contained autodiff, Adam, 400 steps, batch 64, lr 0.02, width d=24, 5 seeds, identical visible ids and external query. Parameter counts are recorded, not forced equal. The baseline implementations are minimal mechanism references in a separate evaluation directory; they are NOT part of the architecture under study and are NOT reproductions of published systems.

W12 (formal withdrawal). An earlier draft reported Fisher p = 0.4998 for 1.0000 versus 0.9987 on a single-hop task, with the attention number taken from a stored record. In this same-run audit the attention references did not converge: across 400 / 1200 / 3000 steps and lr 0.02 / 0.10 they stay in [0.0332, 0.3125], with no arm exceeding 0.3125. The stored number did not reproduce. We withdraw the parity statement entirely and make NO claim of equivalence, superiority or inferiority with respect to attention. These toy references are unoptimised; their low values are a limitation of this audit, not evidence about published models. All rows are released so the audit can be falsified.

4.7 Complete ablation ledger

Every ablation below is a zero-training analytic measurement on the same 512 held-out samples unless stated otherwise; the ACCURACY column always uses the marginal statistic of P5(ii). The only manipulation that breaks the diagonal of Table 5 is removal of the role bit; code choice, gate choice, vocabulary size, length and precision do not.

ablation	setting	acc	statistic / n
code=hadamard	code=hadamard role=True norm=True	1.0000	marginal / 512

code=random	code=random role=True norm=True	1.0000	marginal / 512
code=identity	code=identity role=True norm=True	1.0000	marginal / 512
code=gaussian	code=gaussian role=True norm=True	1.0000	marginal / 512
rho_fast	code=hadamard role=True norm=True	1.0000	marginal / 512
rho_slow	code=hadamard role=True norm=True	1.0000	marginal / 512
w_skew	code=hadamard role=True norm=True	1.0000	marginal / 512
no_norm	code=hadamard role=True norm=False	1.0000	marginal / 512
no_role_mask	code=hadamard role=False norm=True	0.6016	marginal / 512

Table 9 (part 1). Analytic ablations. The role-bit removal row is the single breaking case. A second group varies layout, encoding, gate and duplicate keys. Values for parity/gapped and duplicate keys are the five-seed entries of Table 2 (0.3438 and 0.9328); single-seed values from earlier probes are recorded only in the provenance audit (Table 10) and are not quoted as results. A learned low-rank INTERFACE of rank r >= 5 attains 0.9694 at r=4 and 1.0000 at r=16 over 3 seeds; this is a different experiment from the CODEBOOK Gram-rank collapse of Section 4.4, and the two are not compared. Finally, the adversary permutation that makes the flat distractor peak dominate gives measured accuracy 0.0000 with beta > alpha (0.94950 > 0.82605), while the codebook partition gives 1.0000; this is the concrete witness behind Lemma L1.

4.8 Numeric-provenance audit

Earlier drafts quoted several values for the same configuration. Table 10 records each drift, the single value used in this submission, its statistic, and its source. No number in this paper appears without one of these rows.

quantity	earlier values	value here	statistic / source
parity, gapped	0.2969 (1 seed); 0.2711 (other probe)	0.3438	5-seed mean / Table 2
duplicate keys	0.9316; 0.9414; 0.9328	0.9328	5-seed trained / Table 2
FFT/IIR error	2.27e-16; 2.89e-16; 2.43e-14	<= 3e-16	max over 3 probes / Table 4
chance	0.0313 and 0.25 used together	0.0313	1/V_K, this task only
attention refs	labelled chance-equivalent (withdrawn)	0.0332-0.3125	range, no chance claim / Sec 4.6
fp32 bitwise	claimed then denied	withdrawn W1	real-field only / Appendix B

Table 10. Provenance audit. The value 0.3125 in the attention range is 10/32 and is NOT 1/32; the historical four-pair task has chance 0.25 and is a different task, so 0.25 is never placed beside 0.0313. This is the single numeric convention of the paper.

4.9 Full five-seed paired statistics

Table 12 reports every five-seed configuration used in the adversarial and codebook experiments, with its mean, standard deviation and 10,000-resample paired CI. Entries are marginal accuracies under P5(ii). The paired difference between an explicit role bit and the parity format prior on the gapped layout is +0.6562 with CI [+0.6332, +0.6824] over five seeds, all positive.

mask / layout	mean	std	CI95
explicit / alternating	1.0000	0.0000	[1.0000, 1.0000]
explicit / gapped	1.0000	0.0000	[1.0000, 1.0000]
explicit / dense	1.0000	0.0000	[1.0000, 1.0000]
explicit / gapped	1.0000	0.0000	[1.0000, 1.0000]
parity / gapped	0.3438	0.0314	[0.3176, 0.3668]
none / alternating	0.8621	0.0104	[0.8527, 0.8676]

Table 12. Five-seed statistics. The per-fibre uniform value 0.0625 of Table 6 and the marginal value 0.8621 of this table are different statistics under P5; neither is a correction of the other.

4.10 Reproduction protocol

The anonymised artifact contains one reference implementation of the layer, seven minimal mechanism references used ONLY for the Section 4.6 audit, the reachability enumeration of Section 4.3, and all raw result files. All probes are CPU-only NumPy with no external model weights and complete in minutes on a single core. Seeds are listed per table: 1000+s for training, 7000+s for held-out evaluation, s in (0,..,4) for the five-seed rows; analytic witnesses use fixed seeds recorded with each table. Every number in Tables 2-12 is injected from the raw files at build time by string formatting, so a changed source file changes the PDF and a stale transcription is impossible. A single command regenerates the paper and a second command re-runs all synthetic probes. The verifier checks statement formation, witness agreement and protocol consistency (7/7 gates pass); it is NOT a mathematical proof, and all proofs in this paper are analytic.

Compute statement: total CPU time for every reported synthetic probe is under one hour; no GPU, no pretrained model, no private weight and no external dataset is used. Token counts, perplexity and generation quality are NOT measured. The 2027 call for papers and submission deadline had not been published when this version was prepared; page count, the author-side LLM-use disclosure and reciprocal-reviewer obligations must be rechecked against the official template when it appears.

4.11 Explicit falsification conditions

Each result has a stated failure mode that can be checked without re-running any training. A(b) is falsified by any exact H-hop construction with B < log2|A| on a fibre where |A| > 2^B; the identifiable place is the injection step in Appendix A.1. A(c) is falsified by a duality-preserving layer whose coefficients depend on x_t yet whose impulse response is Toeplitz, contradicting P6's induction on the lag. Result 2(i) is falsified by an order-1 state achieving exact one-hop retrieval under P2; Result 2(ii) by a single static-LTI layer exact on H >= 2 hops, or by a two-layer stack exact on H=1 under P4. P5 is falsified by an exact H-hop construction with o <= H under its d-orthogonal-successor premise. Construction C is falsified by a non-Toeplitz impulse response or an L-dependent state, both checked in Table 4. The W12 audit is falsified by reproducing the stored 0.9987 attention baseline under the Section 4.6 protocol. Each condition names the exact measurement that would have to come out differently.

5 What This Does NOT Establish

· **No parity claim.** W12 withdraws it; the same-run audit did not reproduce the stored baseline. No equivalence, superiority or inferiority is claimed.

· **No architecture comparison.** Section 4.6 supplies same-condition toy mechanism references for a reproduction audit, not full reproductions of GLA, DeltaNet, Mamba-2 or Transformer, and not a performance ranking. Earlier O1-O4 lacked 5-seed paired CIs; the Section 4.6 results have them.

· **No language modeling.** There is no >=10M-token equal-compute run, no val perplexity, no generation readability and no measured throughput comparison.

· **Not a new paradigm.** P4 states the rho=1 single-pole isomorphism with linear attention. The contribution is a tight JOINT characterization under static kernels.

· **Role necessity is conditional.** With no role bit the tight alternating layout gives 0.8621 while a gapped layout gives 0.9609 because zero fillers suppress the spurious peak (L1). Under the codebook partition neither a tag nor a position rule is needed; the alphabet convention remains an encoding-level inductive bias. Exactly one bit per token is a sufficient carrier, not an information-theoretically minimal one.

· **Overshoot is not an architectural law.** Theorem A(f) attributes H_L > H failure to last-layer read-out, not to the static kernel family.

6 Related Work, Placed Precisely

Table 1 gives the exact increment per known result; we do not repeat it here. Zoology [1], Repeat After Me [2] and Illusion of State [3] established that fixed-state models struggle with associative recall. S5 [4] gave the LTI duality and the requirement of time-invariant coefficients; Mamba [5] observed that selectivity breaks the convolution view; linear attention [6] and its gated and delta-rule variants [7,8] use outer-product states. We do not claim to be the first to observe that static convolutions struggle with recall, nor that LTI systems require time-invariance. The increment is the exact algebraic bound B >= log2 |A|, its per-fibre accuracy statement, the tight binary-chain witness, and the joint layer/state characterization under a

strictly static kernel.

6.1 Relation to MQAR and associative-recall protocols

Zoology, MQAR and induction-head tasks measure recall under a large vocabulary, multiple queries and non-adjacent bindings, with performance summarised over a distribution of sequences. The task studied here is deliberately narrower: n_pairs = 4, V_K = 32, one external query, adjacent binding emission and a single last-layer read-out. It was chosen because it admits an EXACT per-fibre answer set A(v) that can be enumerated, which is what makes the bound tight and falsifiable. It is not an MQAR result and is not reported as one. In particular, the 0.9512 five-seed trained accuracy on H=2 (Table 3) is not comparable to an MQAR score: the vocabularies, query counts, sequence lengths and answer distributions differ. What transfers is the information bound A(b), which is distribution-free and therefore applies to MQAR as well: any flat projection that leaves |A(v)| compatible answers requires at least log2|A(v)| input-side bits, regardless of vocabulary size. Establishing the same tightness on a standard MQAR protocol would require enumerating A(v) there, which we did not do and do not claim.

7 Limitations and a Falsifiable Prediction

Theorem A assumes distinct keys, adjacent binding emission, an external query, last-layer read-out and a deterministic decoder; multi-query chains and stop/select mechanisms are outside Result 2. P5's state-order bound is conditional on d mutually orthogonal candidate successors per hop; when the branching factor is 2 it degenerates to 2^(H+1), which we state rather than hide. Finite-precision results cover L <= 128 for the analytic construction; the long-decay frontier at L=2049 is a separate measurement. Theorem A(b) is distribution-free and depth-free: no deterministic decoder supplied with B < log2 |A| input-side bits can be exact on all compatible instances. Any construction attaining exact H-hop retrieval with fewer bits falsifies the bound at a precisely identifiable place. We did not run a >=10M-token equal-compute language-model experiment, and we do not present this as an architecture paper. A learnable codebook, a converged strong baseline and a fair throughput comparison remain open.

8 Conclusion

Exact retrieval needs information that a flat symbol sequence does not contain. We proved a tight lower bound on that information (B >= log2 |A|, Theorem A), stated precisely where the bits may live within the duality-preserving LTI class, and showed that the bit-depth frontier is a right angle: extra layers cannot substitute for missing bits. Two tight structural bounds (order-2 state for one hop; exactly H static-LTI layers for H hops under last-layer read-out) are attained by one Toeplitz, O(1)-state construction, and the layer bound survives end-to-end training. The practical rule is to budget the bits at the input or relax exactness. The earlier attention-parity claim is withdrawn because it did not reproduce (W12). This is an information-bound paper with a constructive counterexample, not a claim of language-model parity, equivalence, superiority or a new SOTA sequence model.

Broader Impact

This is a theoretical and synthetic-experimental study of information limits in fixed-state sequence models. It contains no language-model training, no human-subject data, no deployed system and no generative capability. The immediate practical consequence is a design rule: an input-side information budget must be accounted for before attributing a retrieval failure to model capacity. A possible misuse is to quote the synthetic exactness figures as evidence of general language-model parity, or to treat the withdrawn parity claim as a performance result; both readings are excluded explicitly in Sections 5 and 8. The work does not introduce a new mechanism for surveillance, data collection or large-scale deployment. Its environmental footprint is limited to CPU-only synthetic probes.

References

[1] Arora, S. et al. Zoology: Measuring and Improving Recall in Efficient Language Models. ICLR 2024. arXiv:2312.04927.

[2] Jelassi, S. et al. Repeat After Me: Transformers are Better than State Space Models at Copying. ICML 2024. arXiv:2402.01032.

[3] Merrill, W. et al. The Illusion of State in State-Space Models. ICML 2024. arXiv:2404.08819.

[4] Gu, A. et al. On the Parameterization and Initialization of Diagonal State Space Models (S5). NeurIPS 2022. arXiv:2208.04933.

[5] Gu, A. and Dao, T. Mamba: Linear-Time Sequence Modeling with Selective State Spaces. COLM 2024. arXiv:2312.00752.

[6] Katharopoulos, A. et al. Transformers are RNNs: Fast Autoregressive Transformers with Linear Attention. ICML 2020. arXiv:2006.16236.

[7] Yang, S. et al. Gated Linear Attention Transformers with Hardware-Efficient Training. ICML 2024. arXiv:2312.06635.

[8] Yang, S. et al. Parallelizing Linear Transformers with the Delta Rule over Sequence Length. NeurIPS 2024. arXiv:2406.06484.

[9] Dao, T. and Gu, A. Transformers are SSMs: Generalized Models and Efficient Algorithms Through Structured State Space Duality. ICML 2024. arXiv:2405.21060.

[10] Poli, M. et al. Hyena Hierarchy: Towards Larger Convolutional Language Models. ICML 2023. arXiv:2302.10866.

[11] Romero, D. W. et al. CKConv: Continuous Kernel Convolution For Sequential Data. ICLR 2022. arXiv:2202.06037.

[12] Tay, Y. et al. Long Range Arena: A Benchmark for Efficient Transformers. ICLR 2021. arXiv:2011.04006.

[13] Vaswani, A. et al. Attention Is All You Need. NeurIPS 2017. arXiv:1706.03762.

[14] Olsson, C. et al. In-context Learning and Induction Heads. Transformer Circuits Thread, 2022.

Appendix A Complete Analytic Proofs

All proofs are analytic. Numerical witnesses certify non-vacuity and agreement with the statements; they are not used as premises.

A.1 Theorem A(a) and A(b)

Fix a fibre v. The decoder observes (v, s) with s in (0,1)^B, so F(v, .) has at most 2^B distinct values. Let A(v) be the set of distinct answers over instances compatible with v and in the support. If |A(v)| > 2^B, two instances G, G' in the fibre share (v, s) but have k_H(G) != k_H(G'). F cannot equal both. Under the uniform measure on the fibre, at most 2^B of the |A(v)| answer classes can be correct, hence acc(F | v) <= 2^B / |A(v)|; this proves (a). Exactness on every instance requires |A(v)| <= 2^B for all v, i.e. B >= log2 |A(v)|; this proves (b). Neither step uses P3. The marginal accuracy E_v[acc(F|v)] is NOT bounded by 1/|A| when fibres with |A(v)|=1 carry most of the mass, which is why Section 4 labels every statistic. QED.

Binary-chain witness for |A| = 2^m. Fix m <= n and H = m. At hop h the visible flat sequence contains one key k_h and TWO candidate successor symbols; a bit d_h not recorded in v selects which one is bound, and the selected symbol is the key of hop h+1. Both candidates are present as tokens, so every d in (0,1)^m yields the SAME v and the same q = k_1. Map d to the endpoint id sum_h d_h 2^(h-1); with V_K >= 2^m these are 2^m distinct endpoints. The witness enumerates all 2^m assignments with bit-identical visible inputs (max deviation 0.0e+00). If the pairing is also absent from v, all n! matchings belong to the fibre, giving B >= log2(n!). QED.

A.2 Theorem A(c) and the Toeplitz equivalence (P6)

In class M, layer h computes z_h = sum_t K_h(L-1-t) m_h[t] e(x_t) with m_h[t] = <q_h, e(x_(t-1))>, and q_(h+1) = T_h z_h / ||T_h z_h||. K_h depends only on the lag and on fixed parameters, T_h is fixed, and e is fixed. The composition is therefore a function of the multiset of (flat symbol id, position) pairs plus q, i.e. of v with the refinement s removed. Under arbitrary filler spacing the positions of the pair tokens are determined by filler counts independent of the role assignment, so F is constant on the fibre: no choice of H_L, state order or (rho, theta, w) changes this, and A(a) applies. To encode roles in the kernel one makes the recurrence coefficients functions of x_t, hence time-varying. For y_t = a_t y_(t-1) + b_t y_(t-2) + c_t u_t with c_t != 0, P6 gives h(s,s)=c_s, h(s+1,s)=a_(s+1) c_s, h(s+2,s)=(a_(s+2)a_(s+1)+b_(s+2)) c_s. Toeplitz forces c_s, a_s and b_s constant by induction on the lag, so (rho_t, theta_t) are constant. Non-Toeplitz h(t,s) has no single H(z) and the single-FFT path fails. This negates THAT path within M, not every conceivable O(L log L) algorithm. A(c) is therefore a conditional statement: IF duality is preserved, the bits must enter on the input side. QED.

A.3 Theorem A(d), A(e), A(f) and P2

A(d). By the Section 2 definition B = sum_t log2 |pi^(-1)(v_t)|, so an alphabet partition with |pi^(-1)| = 2 contributes exactly one bit per non-empty token and is COUNTED in B. An explicit tag and a symbol id carrying a role are the same quantity in a different carrier. Tagging only the KEY token of each of n pairs gives B = n, which attains B >= n from A(b). A positional prior is a support restriction: it reduces |A(v)| rather than increasing B, and under random filler spacing the restriction no longer holds, so the effective information falls to zero (measured 0.3438, 5 seeds). QED.

P2 (explicit construction). Fix keys and q; A = sum_p L_p v_p + c with d x d blocks L_p, read-out f(A,q) arbitrary, target j = j(q). Case 1: L_j singular. Pick nonzero delta in ker L_j and change only v_j; delta-A = 0, answer changes. Case 2: L_j invertible and some L_p != 0 for p != j. Pick delta_p and set delta_j = -L_j^(-1) L_p delta_p; then delta-A = L_p delta_p + L_j delta_j = 0, and delta_j is nonzero, so the target value changes. Case 3: L_j invertible and all other L_p = 0. Then A depends only on v_j; fix all values and redirect q to a different pair p != j, which changes the required answer while A is unchanged. In every case two legal assignments give the same (A,q) and different answers, so f cannot be exact. No step infers a target-coordinate perturbation from dim ker L > 0 alone. Scope: single mixer with a fixed linear historical summary; deep stacks with per-position nonlinear FFNs, input-dependent gates and outer-product states leave the premise. QED.

A(e). B < log2|A| makes exactness impossible for EVERY deterministic model by A(b), hence for every H_L; for B >= log2|A| Result 2(ii) gives minimum depth exactly H. The frontier is a right angle. A(f): under P4 the decoded value at depth H_L is k_(H_L); for H_L > H and distinct keys this differs from k_H. Retaining and selecting the H-th intermediate output gives the correct answer for any H_L >= H; the lower bound H_L >= H is unchanged. QED.

A.4 Result 2 (Theorem B): induction and tightness

(i) P2 rules out an order-1 vector state with a linear summary; Construction C gives an order-2 tensor state with a contraction read-out, so the minimum is exactly 2. (ii) Induction on H. Base H=1: one contraction resolves one content match and yields the value; a zero-th layer cannot. Step: suppose H-1 layers resolve H-1 hops and produce q_H = C[k_H] by the unique H6 interface. Layer H performs one further match with q_H and returns k_H's successor. Fewer than H layers leave at least one hop unresolved: the last read-out sees a mixture of at least two successors and cannot be exact on all fibres. The analytic construction attains 1.0000 on the diagonal of the 5 x 5 grid and <= 0.10 off it, so the bound is tight and non-vacuous. Overshoot H_L > H is A(f), not a failure of the lower bound. QED.

A.5 P5: conditional state-order lower bound

Premise. The interface is H6 with rank at most d; the codebook rows are mutually orthogonal; at each of H hops there are d mutually orthogonal candidate successors; answers are determined by the joint choice of the start key and the H successors, so the candidate set has size d^(H+1). Claim. An order-o tensor state whose per-layer linear summary has image dimension at most o*d cannot be exact unless o >= H+1. Proof. The H+1 independent matching directions must be simultaneously recoverable from the summary for every one of the d^(H+1) joint choices. If o <= H, the summary is a linear map from a space of dimension (H+1)d into one of dimension o*d <= H*d; its kernel has dimension at least d > 0. By the P2 construction (with the blocks taken to be the coefficient matrices of the H+1 directions) there exist two distinct joint choices producing the same summary and the same q, hence the same read-out, yet requiring different answers. Exactness is impossible; therefore o >= H+1. This is a CONDITIONAL bound: when each hop offers only 2 candidates the same counting gives 2^(H+1) classes and the necessary order is weaker, degenerating to a log2(H+1) bit requirement. We state that degeneration explicitly rather than presenting d^(H+1) as unconditional. QED.

A.6 P1, P3, P4, P7, P8 and non-vacuity

P1. H(z) = sin(theta) / (1 - 2 rho cos(theta) z^(-1) + rho^2 z^(-2)) gives y_t = 2 rho cos(theta) y_(t-1) - rho^2 y_(t-2) + sin(theta) u_t, a real-field equivalence; different rounding orders in the FFT and IIR paths make fp32 bitwise identity false (W1). P3. For a finite output set F with |F| <= |S|, the state can encode f(x) directly, so the literal claim 'O(1) states must lose' is false; the correct statement is that a SPECTRAL compression state loses content addressing. P4. At rho=1 and a single pole the Construction C state reduces to sum_tau k_tau v_tau^T, the linear-attention state; rho<1 retains a decay spectrum. P7. With impulse response bounded by rho^lag / |sin(theta)| and per-step injection bounded by delta_max, the output error is at most delta_max / ((1-rho)|sin(theta)|), under a uniform floating-point perturbation model. P8. As theta_m -> 0 the dictionary columns approach theta_m rho^lag (lag+1) and become collinear, so the grid is asymptotically rank deficient. Non-vacuity: n=4, m=2, V_K=32, H=2, flat alphabet, arbitrary filler spacing give |A|=4 at B=0 with acc <= 1/4, and exactness at B=2.

Appendix B Withdrawn Statements

ID	Withdrawn statement and reason
W1	Unconditional fp32 bitwise FFT/IIR duality: different rounding orders; real-field equivalence only.
W8	Content addressing requires input-dependent kernel coefficients: refuted by Construction C with strictly static kernels.
W9	Absolute-position parity is a general role basis: a hidden format prior; collapses under arbitrary spacing.

W10/W11	'Exactly one bit per token is necessary and sufficient': the necessary quantity is total B >= log2|A|; one bit per token is a sufficient carrier.
W12	Single-hop attention parity (Fisher p=0.4998): the stored baseline did not reproduce in the same-run audit (Section 4.6). No equivalence or superiority claim is made.

Appendix C Reproducibility

The submission is accompanied by an anonymised artifact containing the reference implementation of the layer, the seven same-run mechanism references used only for the Section 4.6 audit, the reachability enumeration of Section 4.3, and all raw result files. Every number in Tables 2-4 is injected from those files at build time; no value is transcribed by hand. All probes are CPU-only NumPy and complete in minutes. Seeds are listed per table. The environment, dependency versions and a one-command reproduction script are in the artifact. The 2027 call for papers and submission deadline were not published at the time of writing; page count, the author-side LLM-use disclosure and reciprocal-reviewer obligations must be rechecked against the official template when it appears.

No count of statements is presented as a contribution. The withdrawn-statement register in Appendix B is a soundness practice, not an originality claim.